

Data Preprocessing and Visualization: Analyzing Diagnostic Breast Cancer Data

Introduction of Data Set

For this project, I will be reviewing, cleaning, and visualizing a dataset on Breast Cancer Diagnostics from the UCI Machine Learning Repository. This data focuses on information about tumors found in breast cancer patients. The Features are computed from a digitized image of a fine needle aspirate (FNA) of a breast mass. The features describe characteristics of the cell nuclei present in the image such as radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, and fractional dimension. This data was recorded from 569 patients and has 30 features. These features are paired with a diagnosis of either malignant (cancerous) or benign (non-cancerous). The most notable features are mean texture, worst area, and worst smoothness.

Research Goals:

1. Is there a correlation between certain aspects of a tumor and its ultimate diagnosis?
2. Are cancerous (malignant) tumors significantly different from benign tumors in size?
3. How can this data be used to identify commonalities or unique features of a tumor?

```
In [13]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [ ]: pip install ucimlrepo
```

```
In [ ]: from ucimlrepo import fetch_ucirepo

# fetch dataset
breast_cancer_wisconsin_diagnostic = fetch_ucirepo(id=17)

# data (as pandas dataframes)
X = breast_cancer_wisconsin_diagnostic.data.features
y = breast_cancer_wisconsin_diagnostic.data.targets
```

```

# metadata
print(breast_cancer_wisconsin_diagnostic.metadata)

# variable information
print(breast_cancer_wisconsin_diagnostic.variables)

# Import necessary libraries

from sklearn.datasets import load_breast_cancer

breast_cancer = load_breast_cancer()

# Create a DataFrame from the dataset
df = pd.DataFrame(data=breast_cancer.data, columns=breast_cancer.feature_names)

# Add the target column
df['target'] = breast_cancer.target

```

This code chunk displays the first five rows and thirty-five columns of the data.

In [4]: `df.head()`

Out[4]:

	mean radius	mean texture	mean perimeter	mean area	mean smoothness	mean compactness	mean concavity	co
0	17.99	10.38	122.80	1001.0	0.11840	0.27760	0.3001	0
1	20.57	17.77	132.90	1326.0	0.08474	0.07864	0.0869	0
2	19.69	21.25	130.00	1203.0	0.10960	0.15990	0.1974	0
3	11.42	20.38	77.58	386.1	0.14250	0.28390	0.2414	0
4	20.29	14.34	135.10	1297.0	0.10030	0.13280	0.1980	0

5 rows x 31 columns

df.info():

the dataset contains 1 categorical value that must be converted from numerical values.

In [14]: `df.info()`

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 569 entries, 0 to 568
```

```
Data columns (total 31 columns):
```

#	Column	Non-Null Count	Dtype
0	mean radius	569 non-null	float64
1	mean texture	569 non-null	float64
2	mean perimeter	569 non-null	float64
3	mean area	569 non-null	float64
4	mean smoothness	569 non-null	float64
5	mean compactness	569 non-null	float64
6	mean concavity	569 non-null	float64
7	mean concave points	569 non-null	float64
8	mean symmetry	569 non-null	float64
9	mean fractal dimension	569 non-null	float64
10	radius error	569 non-null	float64
11	texture error	569 non-null	float64
12	perimeter error	569 non-null	float64
13	area error	569 non-null	float64
14	smoothness error	569 non-null	float64
15	compactness error	569 non-null	float64
16	concavity error	569 non-null	float64
17	concave points error	569 non-null	float64
18	symmetry error	569 non-null	float64
19	fractal dimension error	569 non-null	float64
20	worst radius	569 non-null	float64
21	worst texture	569 non-null	float64
22	worst perimeter	569 non-null	float64
23	worst area	569 non-null	float64
24	worst smoothness	569 non-null	float64
25	worst compactness	569 non-null	float64
26	worst concavity	569 non-null	float64
27	worst concave points	569 non-null	float64
28	worst symmetry	569 non-null	float64
29	worst fractal dimension	569 non-null	float64
30	target	569 non-null	int64

```
dtypes: float64(30), int64(1)
```

```
memory usage: 137.9 KB
```

```
In [ ]:
```

```
In [15]: df.isnull().sum()
```

```

Out[15]: mean radius      0
         mean texture    0
         mean perimeter   0
         mean area        0
         mean smoothness  0
         mean compactness 0
         mean concavity    0
         mean concave points 0
         mean symmetry     0
         mean fractal dimension 0
         radius error      0
         texture error     0
         perimeter error   0
         area error        0
         smoothness error  0
         compactness error 0
         concavity error   0
         concave points error 0
         symmetry error    0
         fractal dimension error 0
         worst radius      0
         worst texture     0
         worst perimeter   0
         worst area        0
         worst smoothness  0
         worst compactness 0
         worst concavity    0
         worst concave points 0
         worst symmetry     0
         worst fractal dimension 0
         target            0
         dtype: int64

```

In []:

The dataset contains no missing values.

```

In [19]: Diagnosis = ['target']
         df[Diagnosis] = df[Diagnosis].astype('category')

```

```

In [20]: diagnosis_count = df['target'].value_counts()
         diagnosis_count

```

```

Out[20]: target
         1    357
         0    212
         Name: count, dtype: int64

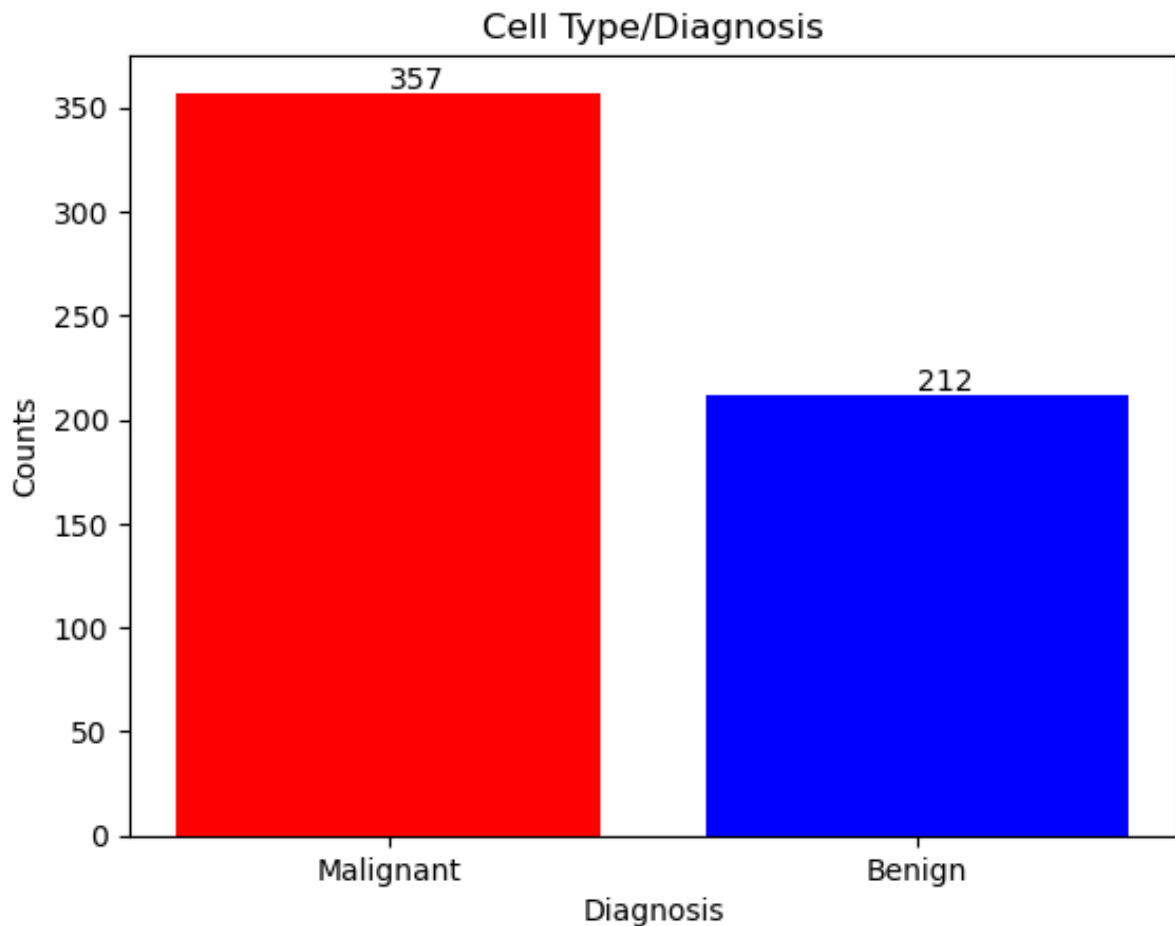
```

This tells us that in this dataset, 357 cases were malignant tumors and 212 were benign.

```
In [21]: diagnosis_type = ['Malignant', 'Benign']
counts = diagnosis_count.values
colors = ['red', 'blue']
```

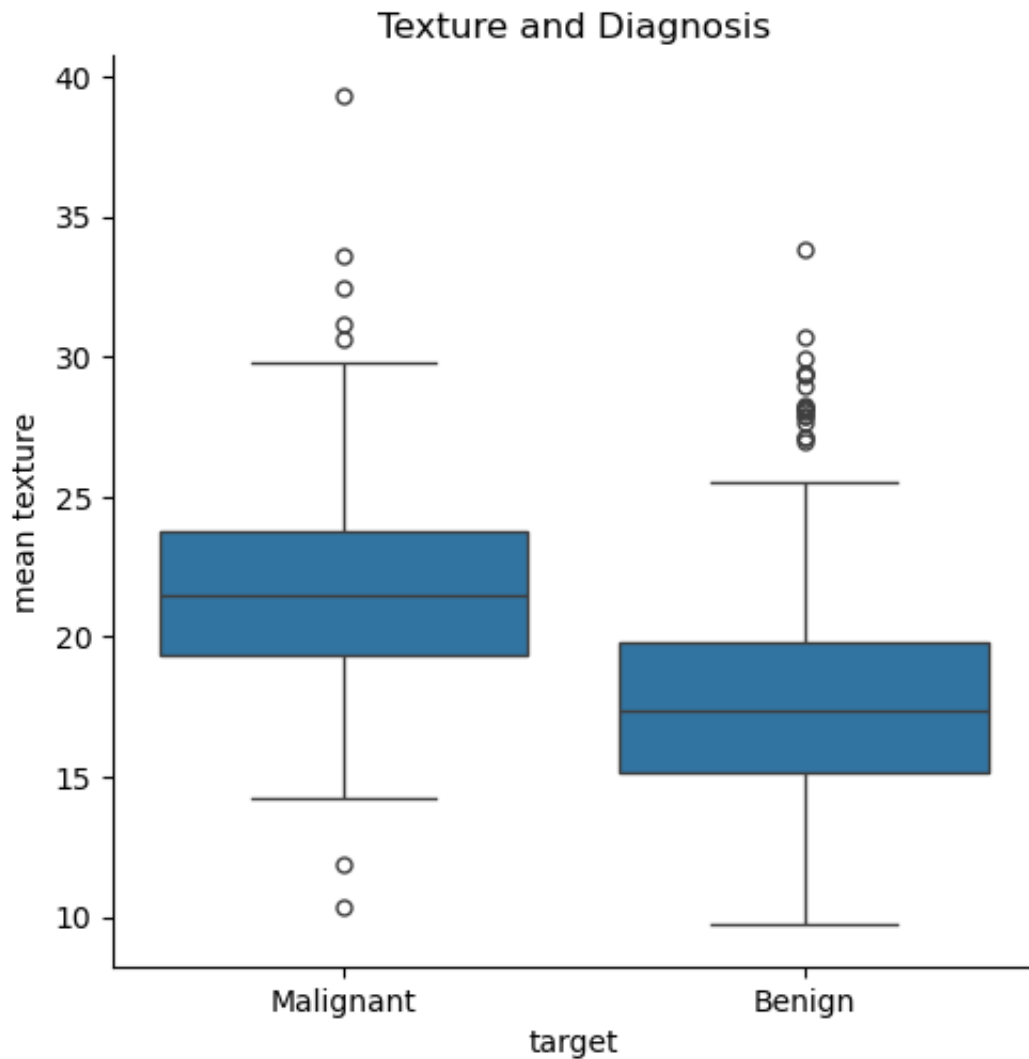
```
In [10]: plt.figure
bars = plt.bar(x = diagnosis_type,
               height = counts,
               color = colors)
for bar in bars:
    height = bar.get_height()
    plt.text(bar.get_x() + bar.get_width()/2,
             height+2,
             str(height),)
plt.xlabel('Diagnosis')
plt.ylabel('Counts')
plt.title('Cell Type/Diagnosis')
plt.show
```

```
Out[10]: <function matplotlib.pyplot.show(close=None, block=None)>
```



```
In [26]: df['target'] = df['target'].map({0: 'Malignant', 1: 'Benign'}) # so instead
```

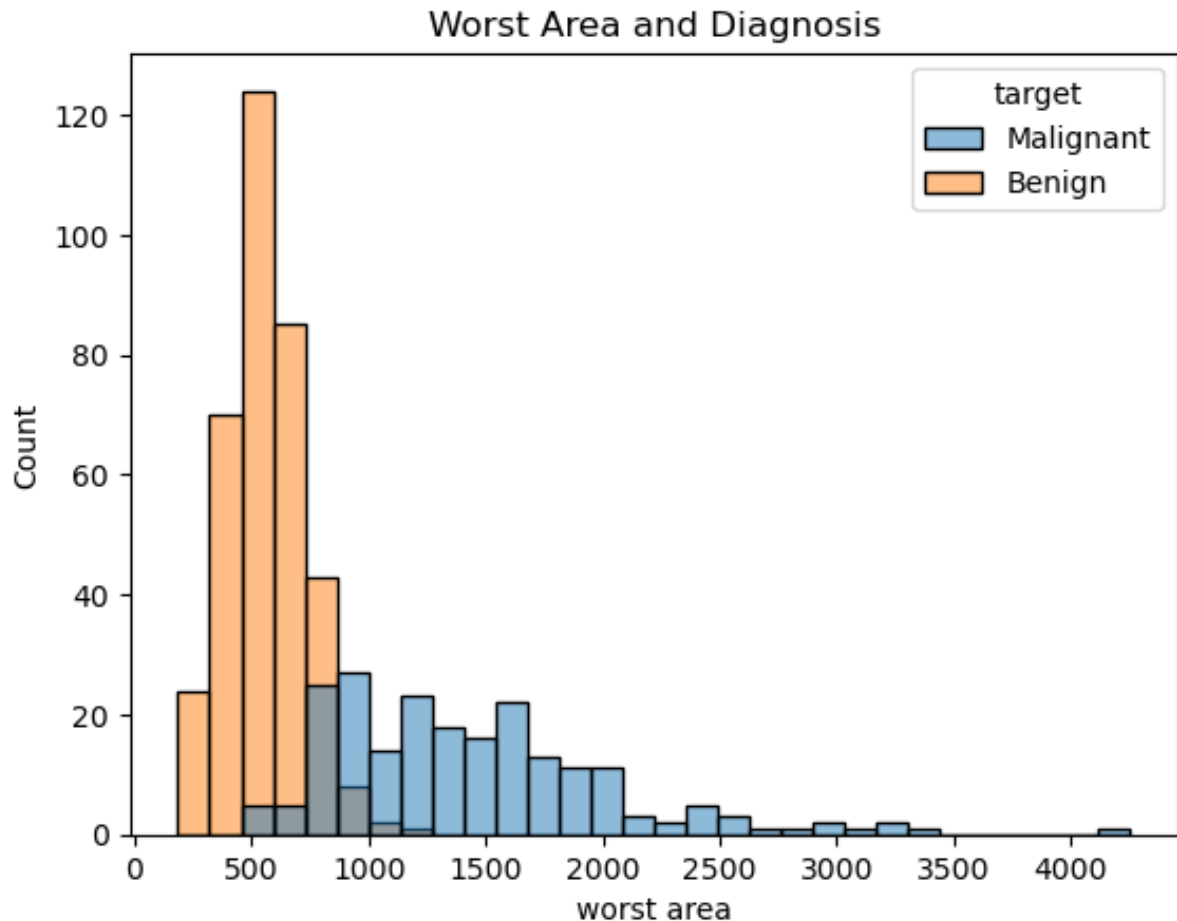
```
In [31]: sns.catplot(x = 'target', y = 'mean texture', data = df, kind = 'box')
plt.title('Texture and Diagnosis')
plt.show()
```



What This Boxplot shows:

We can observe that the median for malignant tumors is higher than the median of benign. What this suggests is there may be an association between tumor texture and cancer.

```
In [29]: sns.histplot(data = df, x = 'worst area', hue = 'target')
plt.title("Worst Area and Diagnosis")
plt.show()
```



In []:

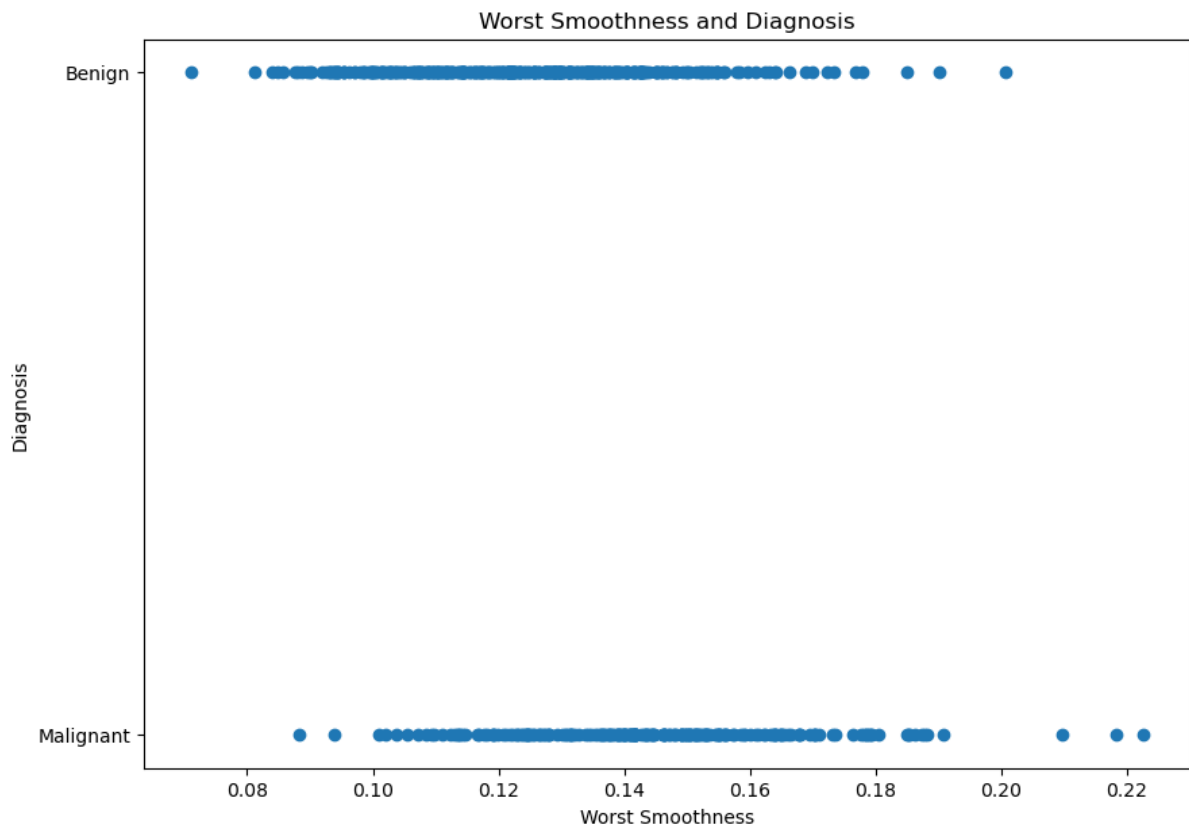
In []:

In []:

What this Graph shows

We can observe that the area for malignant tumors appears to be larger for most cases in comparison to benign tumors, which the majority of are smaller. This tells us that larger tumor size may indicate a higher likelihood of being cancerous.

```
In [34]: plt.figure(figsize = (10, 7))
x = df['worst smoothness']
y = df['target']
plt.plot(x, y, 'o')
plt.xlabel('Worst Smoothness')
plt.ylabel('Diagnosis')
plt.title('Worst Smoothness and Diagnosis')
worst_smoothness = plt.show()
```



What this Scatterplot shows

We can observe that there is a slight cluster pattern for each diagnosis. The malignant tumors tend to cluster around a smoothness of roughly .14, while the benign tumors cluster closer to .12. This information suggests that the smoothness (variation in radius lengths) has more variation or is more uneven in malignant tumors.

Project Summary:

I am interested in this data set because breast cancer is a common disease that affects many people. I also am interested in the healthcare sector because it provides an opportunity to help people. Being able to predict or even better understand cancer and how we can diagnose it is so important because of the impact it could have on the lives of those affected. My first research question was, is there a correlation between certain aspects of a tumor and its ultimate diagnosis? After reviewing the data and visualizations, we can see that there are relationships between the variables of interest and the diagnoses. We saw that according to the graphs, malignant tumors tend to be larger, more textured, and less even in comparison with benign tumors. My second research question was, Are cancerous (malignant) tumors significantly different from benign tumors in

size? The histogram of worst area shows that malignant tumors are very often larger in size than benign tumors. From this we can gather that size is a contributing factor to determining the state of a tumor. My third research question was, how can this data be used to identify commonalities or unique features of a tumor? We can gather information from the plots and graphs to conclude that malignant tumors often have larger area and are more irregular. With that being said, we can also conclude that benign tumors are typically smaller in size and have a smoother and more even texture with fewer irregularities. This helps us to potentially classify tumors based on these variables.

This project is important to both society and business. To society, it is important because it can help improve the accuracy of a diagnosis, or increase the speed that cancer can be detected which is crucial. By using these tools, doctors can be assisted by data in making decisions. In terms of business, this could help save time and money by making a diagnosis faster and more efficient. This could help reduce costs for healthcare companies, hospitals, and patients. To connect mindfulness to this project, I think it is important to remember that it is not just numbers that are being looked at, but patients who are real people. It is important to be mindful of the fact that visualizing this data could help people which makes it that much more worth while. The next step I would take with this data set would be to take a closer look at other variables like compactness or concavity and see if there is a relationship with the diagnosis. Something else I would consider with this data set is delving even deeper into the variables of interest and creating a test for accuracy by using the patterns found in the data.

In []: